

## MAIQUETIA, Venezuela

WMO: 804150

Lat: 10.603N

Lon: 66.991W

Elev: 72

StdP: 100.47

Time Zone: -4.00 (W04)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 2             | 20.8       | 21.8 | 15.8                          | 11.3 | 25.0 | 16.9 | 12.1 | 24.7 | 5.1                   | 27.7 | 4.2 | 28.4 | 0.5                   | 90   | 0.294 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 9             | 7.1                    | 33.9            | 27.8 | 33.1 | 27.4 | 32.8 | 27.3 | 29.4                | 32.1 | 28.8 | 31.6 | 28.2 | 31.1 | 2.6                  | 0    |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 28.9                            | 25.7 | 31.2 | 28.1 | 24.4 | 30.9 | 27.2 | 23.2 | 30.3 | 96.9          | 32.5 | 94.5 | 31.4 | 91.4 | 31.6 | 32.0              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |      |     |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|------|-----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |      |     |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |      |     |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |      |     |
| (4) 4.2           | 3.4  | 3.2 | DB                         | 20.0 | 36.5               | 1.0 | 1.6                                                | 19.3 | 37.7       | 18.7 | 38.6       | 18.1 | 39.5       | 17.4 | 40.6 | (4) |
| (5)               |      |     | WB                         | 16.6 | 30.2               | 1.0 | 0.9                                                | 15.9 | 30.8       | 15.3 | 31.4       | 14.8 | 31.9       | 14.1 | 32.5 | (5) |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |  |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|--|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 27.7        | 26.3  | 26.2  | 26.6  | 27.3  | 28.0  | 28.2  | 28.0  | 28.6  | 28.9  | 28.6  | 28.1  | 27.1  | (6)  |  |
|                                                                        | DBStd       | 1.50        | 1.07  | 1.34  | 1.24  | 1.20  | 1.20  | 1.26  | 1.35  | 1.37  | 1.32  | 1.13  | 1.00  | 0.90  | (7)  |  |
|                                                                        | HDD10.0     | 0           | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | (8)  |  |
|                                                                        | HDD18.3     | 0           | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | (9)  |  |
|                                                                        | CDD10.0     | 6449        | 504   | 453   | 516   | 519   | 558   | 547   | 559   | 575   | 568   | 577   | 542   | 531   | (10) |  |
|                                                                        | CDD18.3     | 3408        | 246   | 220   | 258   | 269   | 299   | 297   | 301   | 317   | 318   | 319   | 292   | 273   | (11) |  |
|                                                                        | CDH23.3     | 34214       | 1960  | 1762  | 2206  | 2554  | 3114  | 3153  | 3162  | 3552  | 3706  | 3546  | 3021  | 2478  | (12) |  |
|                                                                        | CDH26.7     | 11784       | 539   | 487   | 626   | 767   | 1049  | 1121  | 1125  | 1389  | 1532  | 1367  | 1045  | 736   | (13) |  |
| (14) Wind                                                              | WSAvg       | 1.1         | 1.0   | 1.1   | 1.2   | 1.2   | 1.2   | 1.1   | 1.1   | 1.1   | 1.1   | 1.0   | 0.9   | 1.0   | (14) |  |
| (15) Precipitation                                                     | PrecAvg     | 504         | 29    | 23    | 17    | 25    | 32    | 49    | 58    | 52    | 49    | 47    | 65    | 49    | (15) |  |
|                                                                        | PrecMax     | 1024        | 203   | 148   | 70    | 253   | 105   | 183   | 132   | 200   | 135   | 153   | 429   | 193   | (16) |  |
|                                                                        | PrecMin     | 142         | 0     | 0     | 0     | 0     | 0     | 0     | 6     | 1     | 6     | 2     | 0     | 0     | (17) |  |
|                                                                        | PrecStd     | 167         | 40    | 30    | 21    | 41    | 26    | 34    | 31    | 40    | 31    | 38    | 78    | 51    | (18) |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 31.9  | 32.8  | 32.2  | 32.8  | 33.7  | 34.0  | 33.2  | 34.2  | 35.0  | 34.1  | 33.2  | 32.1  | (19) |  |
|                                                                        |             | MCWB        | 26.7  | 27.4  | 27.6  | 27.6  | 27.0  | 27.1  | 27.0  | 27.9  | 28.6  | 28.0  | 27.5  | 25.9  | (20) |  |
|                                                                        | 2%          | DB          | 30.9  | 31.0  | 31.1  | 31.8  | 32.2  | 32.8  | 32.3  | 33.1  | 33.9  | 33.1  | 32.3  | 31.2  | (21) |  |
|                                                                        |             | MCWB        | 25.9  | 25.9  | 26.1  | 26.8  | 27.0  | 27.0  | 26.7  | 27.5  | 28.0  | 27.6  | 27.1  | 25.9  | (22) |  |
|                                                                        | 5%          | DB          | 30.1  | 30.1  | 30.2  | 31.0  | 31.5  | 32.0  | 31.9  | 32.4  | 33.0  | 32.6  | 31.9  | 30.8  | (23) |  |
|                                                                        |             | MCWB        | 25.2  | 25.3  | 25.5  | 26.3  | 26.7  | 26.5  | 26.6  | 27.0  | 27.5  | 27.2  | 26.9  | 25.8  | (24) |  |
|                                                                        | 10%         | DB          | 29.2  | 29.2  | 29.8  | 30.1  | 30.9  | 31.1  | 31.1  | 31.9  | 32.1  | 31.9  | 31.0  | 30.0  | (25) |  |
|                                                                        |             | MCWB        | 24.7  | 24.7  | 25.3  | 25.8  | 26.4  | 26.4  | 26.3  | 26.7  | 27.0  | 26.8  | 26.4  | 25.4  | (26) |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 27.8  | 28.5  | 29.1  | 29.3  | 28.8  | 28.9  | 29.1  | 29.8  | 30.4  | 29.8  | 29.2  | 28.2  | (27) |  |
|                                                                        |             | MCDB        | 30.5  | 31.1  | 30.7  | 30.9  | 31.4  | 31.8  | 32.0  | 33.1  | 33.1  | 32.7  | 32.0  | 30.5  | (28) |  |
|                                                                        | 2%          | WB          | 26.6  | 27.0  | 27.8  | 28.0  | 28.0  | 28.0  | 28.1  | 28.9  | 29.2  | 28.6  | 28.2  | 27.2  | (29) |  |
|                                                                        |             | MCDB        | 29.5  | 29.8  | 29.8  | 29.7  | 30.8  | 31.0  | 31.0  | 31.9  | 32.2  | 31.9  | 31.2  | 29.9  | (30) |  |
|                                                                        | 5%          | WB          | 26.0  | 26.1  | 26.5  | 27.1  | 27.4  | 27.4  | 27.4  | 28.1  | 28.5  | 27.9  | 27.6  | 26.6  | (31) |  |
|                                                                        |             | MCDB        | 29.0  | 28.9  | 29.2  | 29.6  | 30.2  | 30.4  | 30.5  | 31.0  | 31.6  | 31.3  | 30.6  | 29.4  | (32) |  |
|                                                                        | 10%         | WB          | 25.4  | 25.5  | 25.8  | 26.4  | 26.8  | 26.9  | 26.7  | 27.3  | 27.7  | 27.3  | 26.9  | 26.1  | (33) |  |
|                                                                        |             | MCDB        | 28.5  | 28.4  | 28.7  | 29.1  | 29.7  | 29.9  | 29.9  | 30.4  | 30.9  | 30.7  | 29.9  | 28.8  | (34) |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 7.1   | 6.9   | 6.6   | 6.3   | 6.3   | 6.5   | 6.6   | 6.9   | 7.1   | 6.9   | 6.8   | 6.8   | (35) |  |
|                                                                        |             | MCDBR       | 7.5   | 7.3   | 6.9   | 6.6   | 6.8   | 7.0   | 7.0   | 7.3   | 7.6   | 7.4   | 7.4   | 7.3   | (36) |  |
|                                                                        | 5% WB       | MCWBR       | 5.4   | 5.3   | 5.0   | 4.4   | 4.1   | 4.2   | 4.5   | 4.9   | 4.9   | 4.9   | 4.9   | 5.1   | (37) |  |
|                                                                        |             | MCWBR       | 7.3   | 7.1   | 6.7   | 6.4   | 6.6   | 6.8   | 6.8   | 7.0   | 7.3   | 7.2   | 7.1   | 6.9   | (38) |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        | taub        | 0.400 | 0.430 | 0.439 | 0.516 | 0.493 | 0.517 | 0.493 | 0.461 | 0.435 | 0.416 | 0.404 | 0.414 | (40) |  |
|                                                                        |             | taud        | 2.483 | 2.372 | 2.365 | 2.120 | 2.231 | 2.138 | 2.222 | 2.346 | 2.419 | 2.490 | 2.524 | 2.471 | (41) |  |
|                                                                        | Ebn at Noon | Ebn at Noon | 907   | 890   | 883   | 808   | 812   | 782   | 804   | 842   | 874   | 891   | 897   | 885   | (42) |  |
|                                                                        |             | Edh at Noon | 109   | 125   | 128   | 163   | 142   | 154   | 142   | 128   | 120   | 110   | 104   | 109   | (43) |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      | RadAvg      | 5.27  | 5.78  | 5.96  | 5.70  | 5.52  | 5.63  | 5.73  | 5.90  | 5.87  | 5.43  | 4.91  | 4.88  | (44) |  |
|                                                                        | RadStd      | RadStd      | 0.24  | 0.31  | 0.44  | 0.38  | 0.30  | 0.21  | 0.16  | 0.20  | 0.19  | 0.19  | 0.38  | 0.26  | (45) |  |

## Historical Trends

|                       | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|-----------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                       |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                       |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)     | (k)     | (l)     |
| Station Only          | +0.78 | N/A     | N/A    | +0.55   | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| Regional (1 neighbor) | +0.78 | N/A     | N/A    | +0.55   | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page